

Learning Quantitative Automata Modulo Theories

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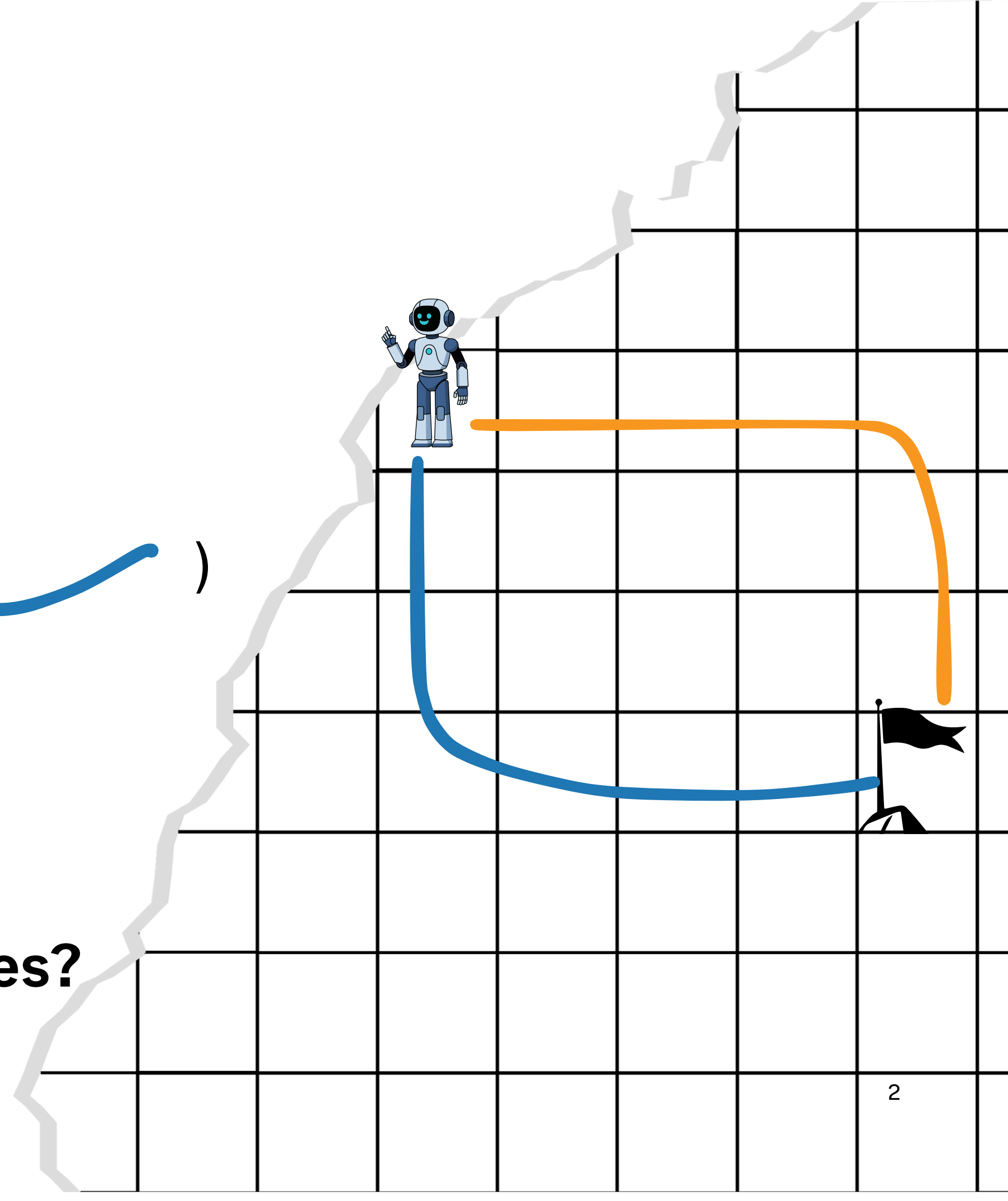
The University of Texas at Austin

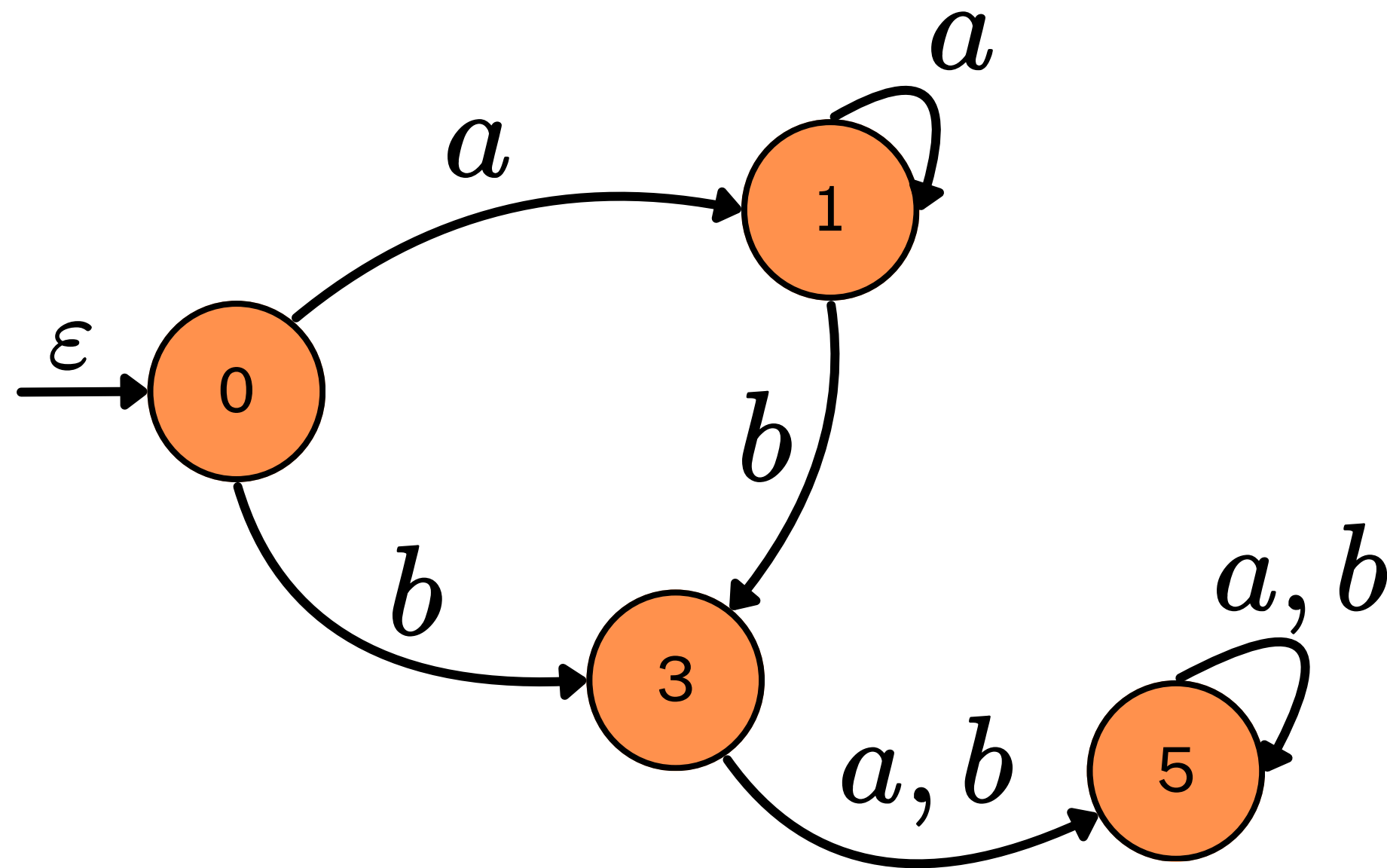
Sequential Decision Making

Objective function

Total Cost() > Total Cost()

What about history-based objectives?





Summation Valuation

$$V(\varepsilon) = 0$$

$$V(a) = 0 + 1 = 1$$

$$V(b) = 0 + 3 = 3$$

$$V(aa) = 0 + 1 + 1 = 2$$

$$V(ab) = 0 + 1 + 3 = 4$$

$$V(ba) = 0 + 3 + 5 = 8$$

$$V(bb) = 0 + 3 + 5 = 8$$

$$V(aaa) = 3$$

Aggregate state labels

$$V(aba) = 9$$

$$V(abb) = 9$$

$$V(baa) = 13$$

...

Input Alphabet $\{a, b\}$

Output Alphabet $\{0, 1, 3, 5\}$

Explicitly specifying a quantitative automaton objective can be challenging

How can quantitative automata be learned from valuation-based preferences?

Preferences (Valuation-based)

$$\varepsilon \prec a$$

$$\varepsilon \prec b$$

$$b \succ a$$

$$b \succ aa$$

$$a \prec aa$$

$$aa \prec ab$$

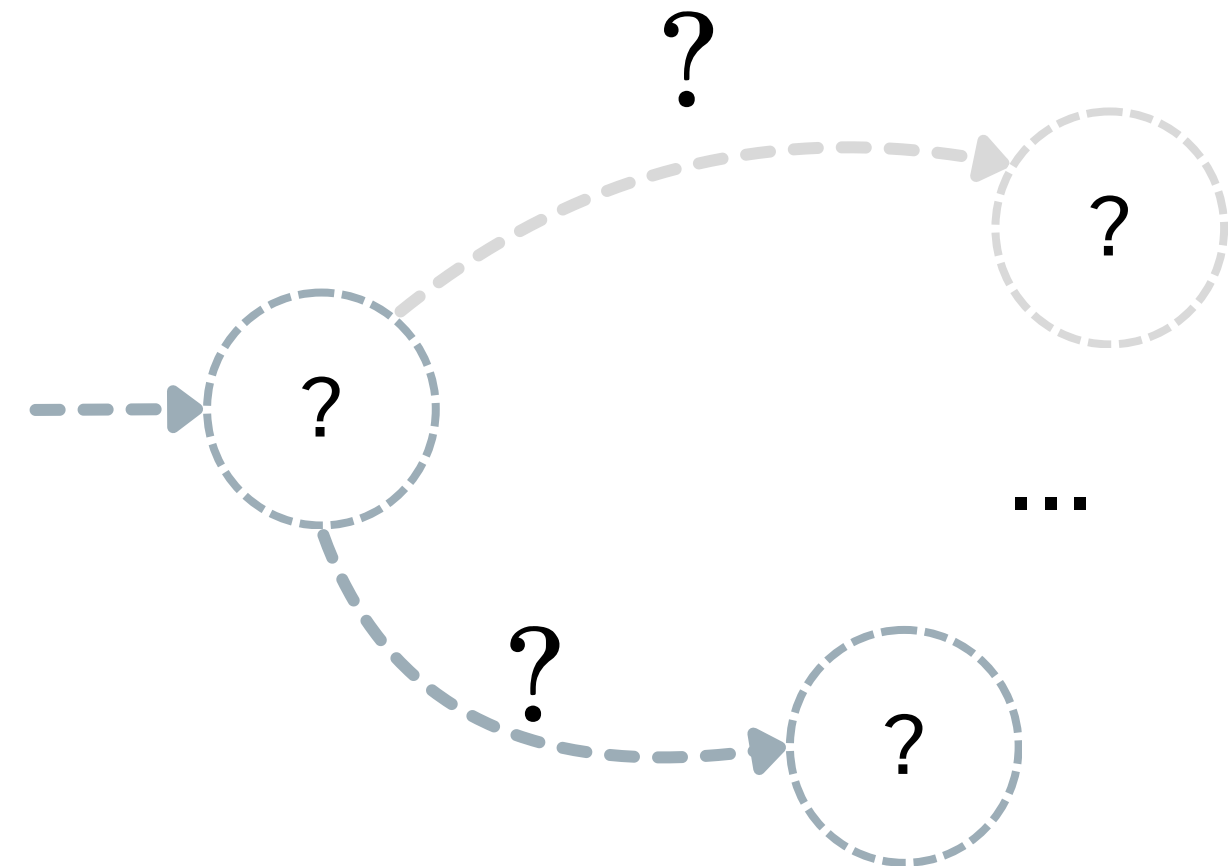
$$ba \sim bb$$

...

Valuation Function

V

What is the structure of the automaton?



States? Transitions? Labels?

Ambiguity is not handled by existing methods

- **Valuation-based preferences** impose significant **ambiguity**, even if preferences are self-consistent:

Growing system of Inequalities

$\varepsilon \prec a$	$V(\varepsilon) < V(a)$
$\varepsilon \prec b$	$V(\varepsilon) < V(b)$
$b \succ a$	$V(b) > V(a)$
$b \succ aa$	$V(b) > V(aa)$
$a \prec aa$	$V(a) < V(aa)$
$aa \prec ab$	$V(aa) < V(ab)$
$ba \sim bb$	$V(ba) = V(bb)$
$\dots$	$\dots$

- Existing preference-based automata learning methods cannot handle such ambiguity.

Ambiguity is not handled by existing methods

- Valuation-based preferences impose significant **ambiguity**, even if preferences are self-consistent:

Example: If a, b, c, d are real numbers, and $a + b > c + d$, how many **feasible** ways can a, b, c , and d be ordered? (Equalities are allowed).

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Example: If a, b, c, d are real numbers, and $a + b > c + d$, how many **feasible** ways can a, b, c , and d be ordered? (Equalities are allowed).

Number of Distinct Values (# Equivalence Classes of Variables)	Number of Total Preorders (unconstrained)	Number of Feasible
4 $\{a\}\{b\}\{c\}\{d\}$	24	16
3 $\{a\}\{b, c\}\{d\}$	36	20
2 $\{a, b\}\{c, d\}$	14	5
1 $\{a, b, c, d\}$	1	0

A system of inequalities has many satisfying **equivalence classes of variables**.
****Identifying states and transitions relies on identifying equivalence classes.****

QUINTIC

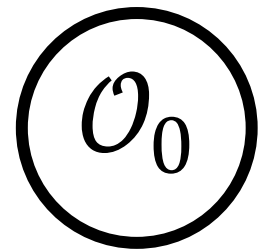
QUINTIC conjectures equivalence classes of variables
within an active learning algorithm

A **sequence of conjectures** (eventually) maps to a **hypothesis automaton**
{states, transitions, labels}

QUINTIC

Tree of conjectures

Representation
and **Constraints**
from **Preferences**



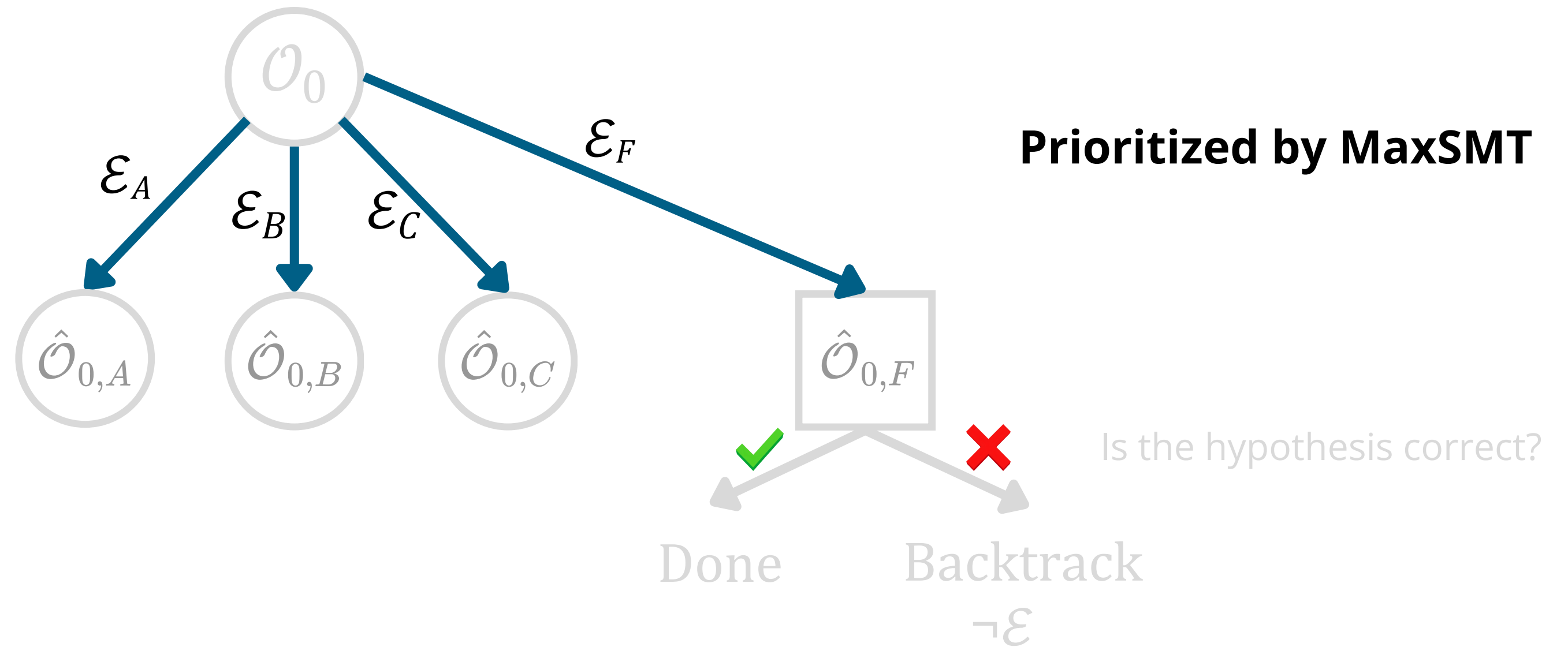
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Conjecture equivalences classes of variables

Representation and
Constraints from
Preferences

Conjectures

Subsequent
Representation



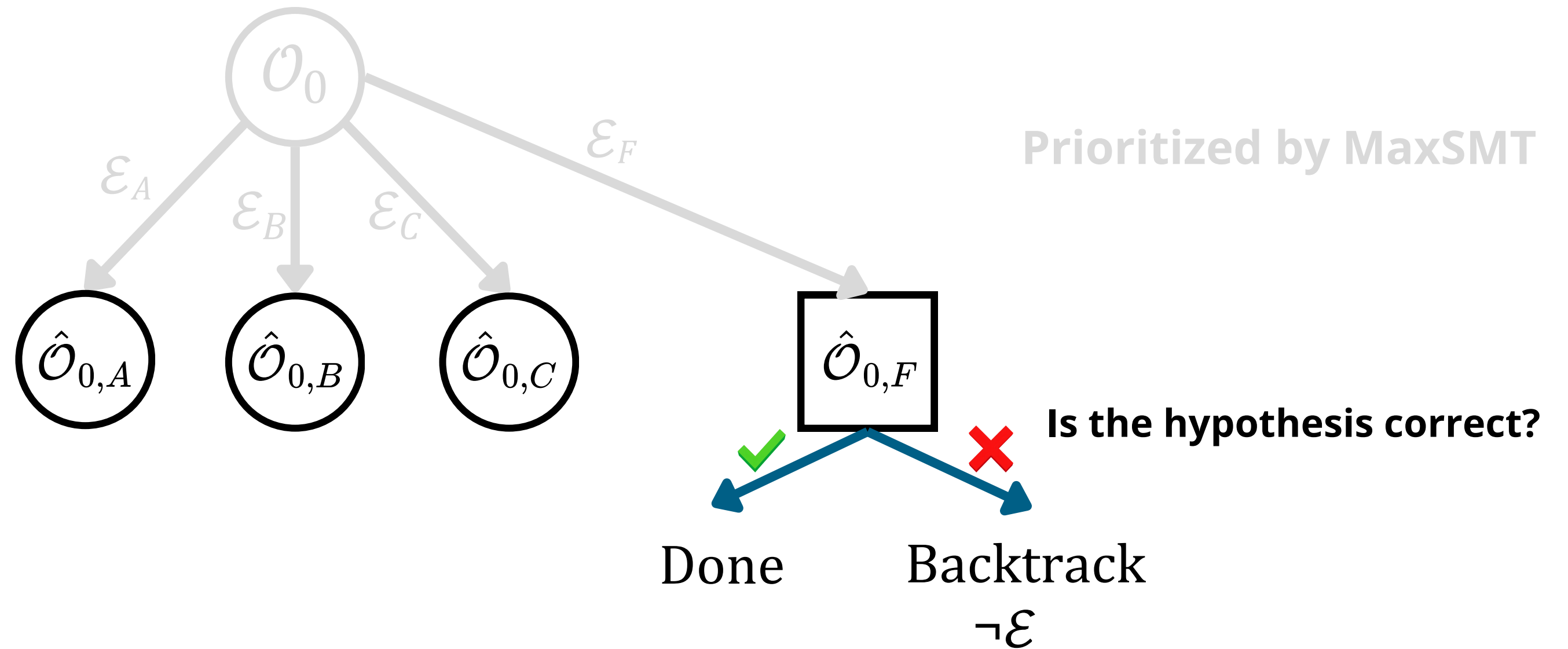
QUINTIC

- Check validity of conjectures given subsequent constraints
- Construct hypothesis and test validity at certain nodes

Representation and
Constraints from
Preferences

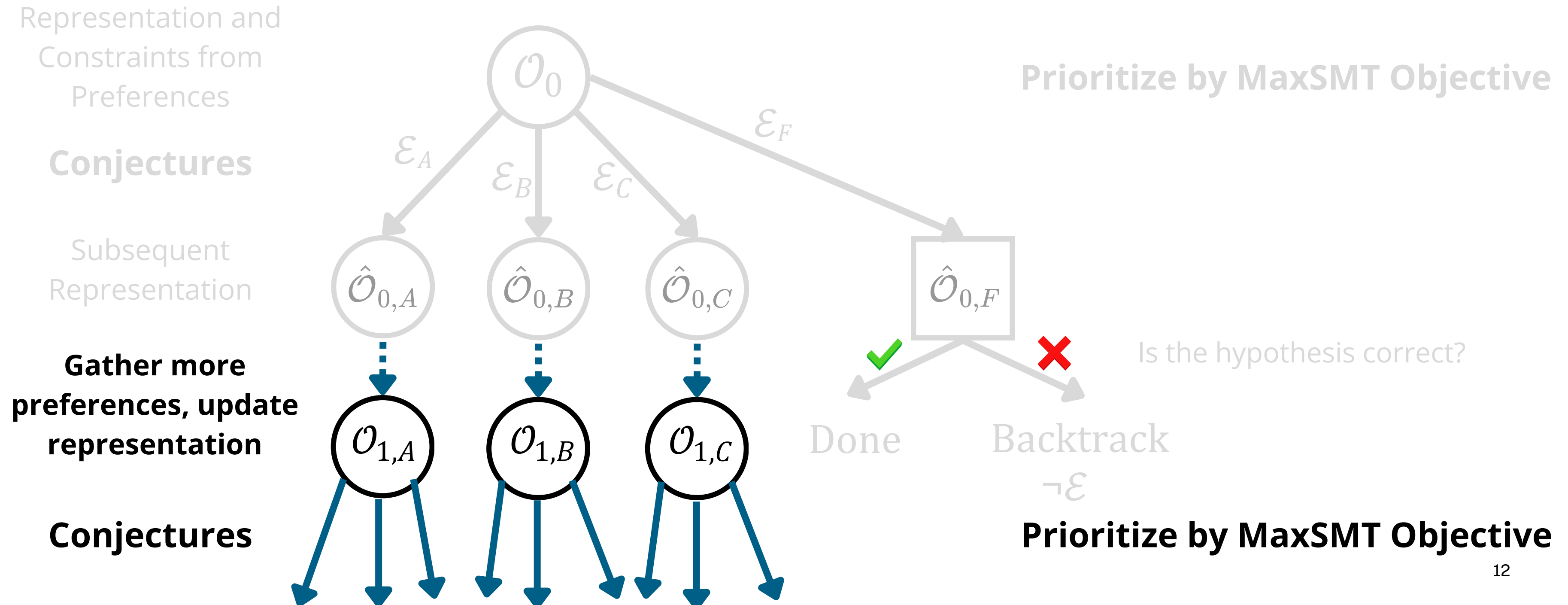
Conjectures

**Subsequent
Representation**



QUINTIC

- Check validity of conjectures given subsequent constraints
- Construct hypothesis and test validity at certain nodes



QUINTIC Guarantees

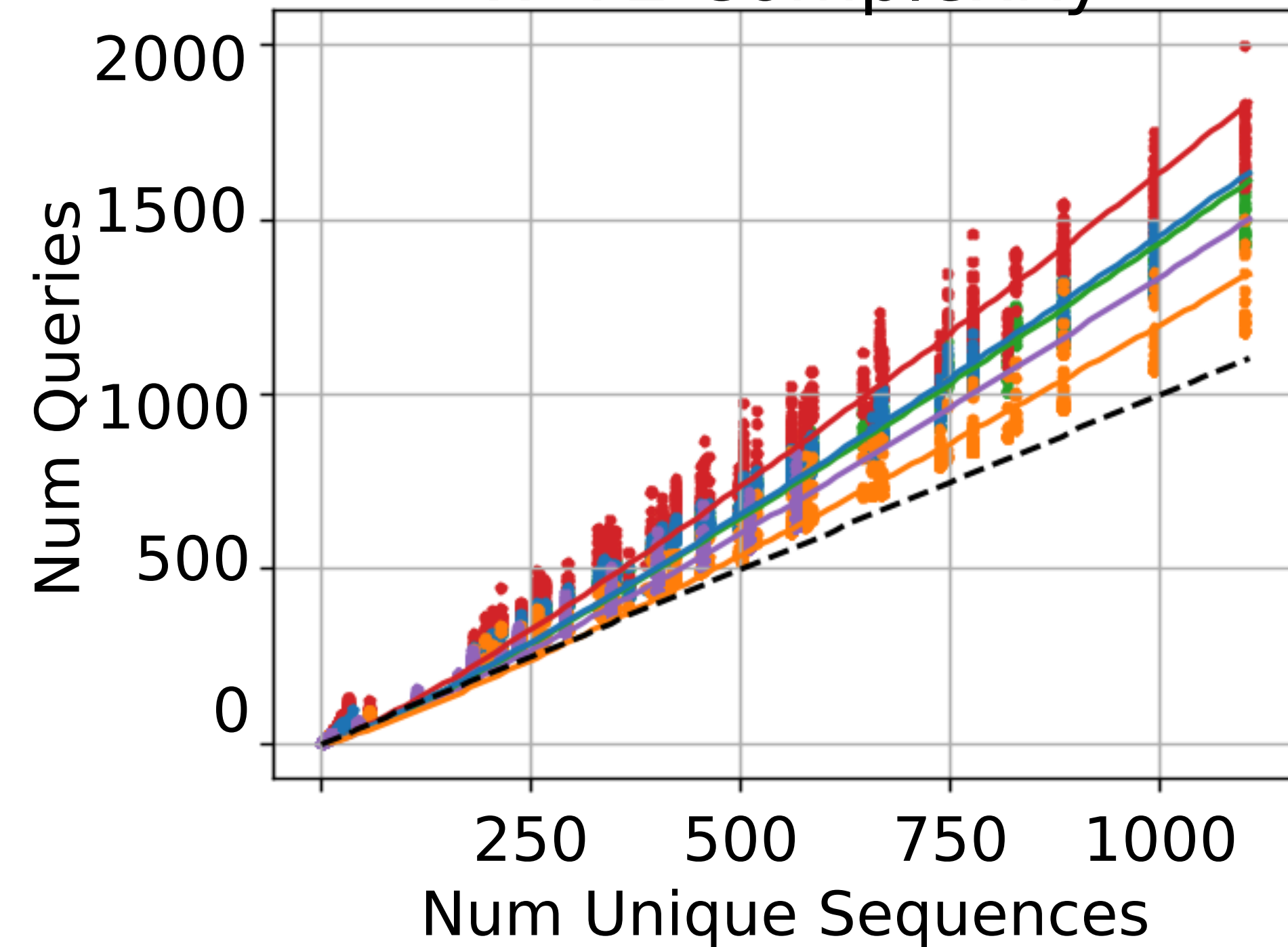
Minimalism

Completeness

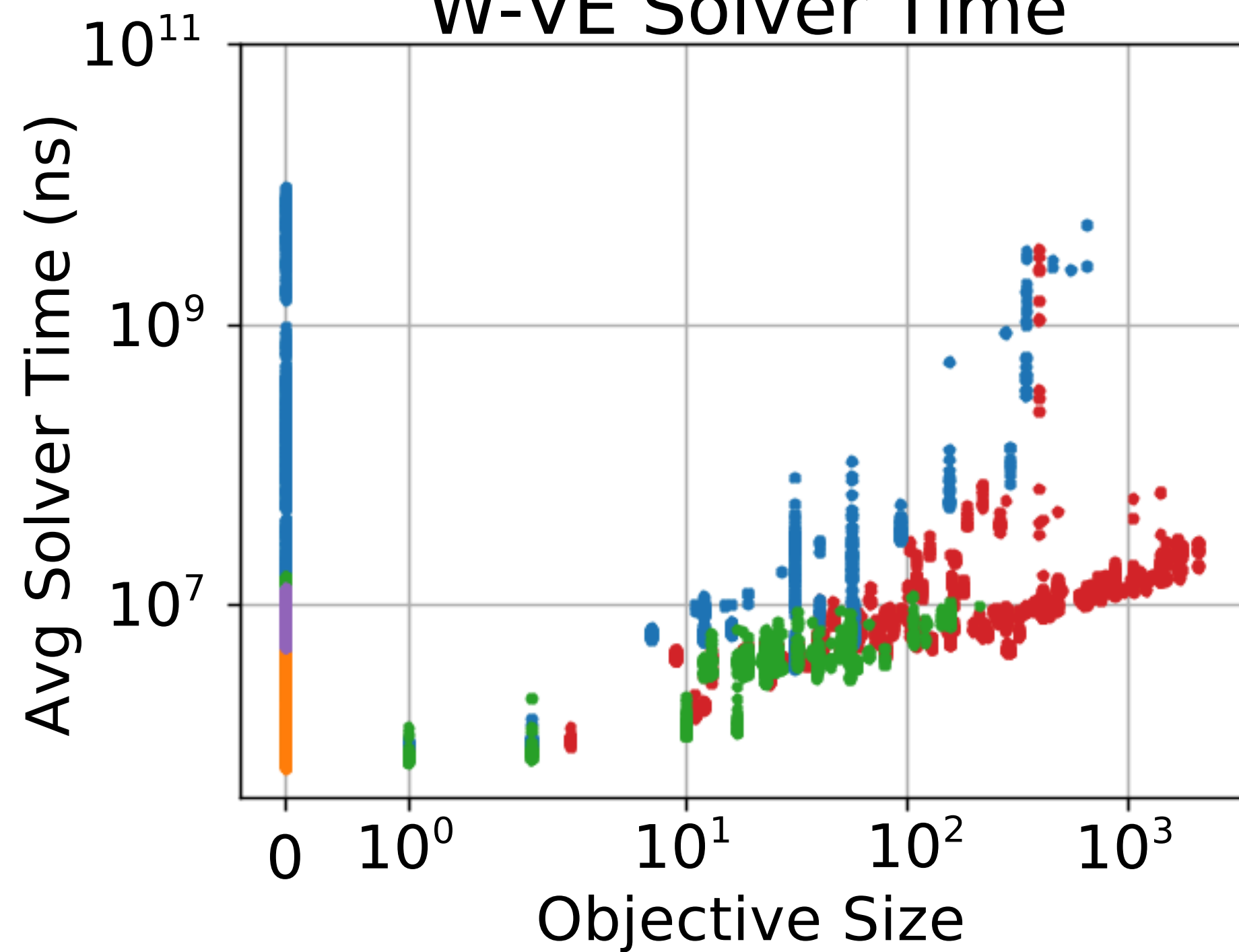
Upper Bounds on Query Complexity

Σ Π REMAP
 $\gamma\Sigma$ $\approx$

W-VE Complexity



W-VE Solver Time



Conclusion

QUINTIC:

- Preference-based active learning algorithm
- Uses MaxSMT to prioritize conjectures of equivalence classes of variables
- Theoretical guarantees of minimalism, completeness, and bounds on query complexity

